

Toy Haven

Assignment 3 — COMP40053 — front-end website submission document

Live website	https://nihmathullahmazhar.github.io/toy-haven/
GitHub repository	https://github.com/nihmathullahmazhar/toy-haven
Technologies	HTML5, CSS3 (Flexbox and Grid, custom properties), vanilla JavaScript (ES5-compatible syntax, no libraries or frameworks)
Storage	<code>localStorage</code> — basket, wishlist, order history, feedback messages, newsletter subscribers, theme preference
Pages	Home, Product listing, Shopping cart, Checkout, Wishlist / collection, Feedback & support (plus an offline fallback page)

1. Overview

Toy Haven is a multi-page front-end store for collectible figurines, toys, board games and diecast model cars. It was built mobile-first with no frameworks: three organised stylesheets, one shared JavaScript core and one small script per page. The 24-product catalogue is a JSON array in `js/data.js`, so every listing, card, modal, basket row and order line is rendered from a single source of data.

Because there is no back end, every interaction that a real store would send to a server is instead written to `localStorage`: the basket survives a page refresh and a browser restart, the wishlist keeps a status per product, checkout writes an order into a local history, and the feedback form stores each message so the visitor can see exactly what was captured. The site is also a progressive web app — it can be installed to a home screen and previously visited pages keep working with no connection.

2. Planning and design

2.1 Approach

Every page was wireframed at 375px first and then widened, which is why the CSS contains only `min-width` media queries: the mobile layout is the default and larger screens add columns

rather than remove them. The breakpoints used throughout are **40em (640px)** for small tablets, **48em (768px)** for the modal and footer, **55em (880px)** where the navigation collapses into the animated hamburger, and **60em (960px)** for the full desktop layout.

2.2 Colour palette and typography

Design tokens — declared once in `css/base.css` and reused everywhere

Token	Value	Used for
<code>--brand</code>	#5b2bd9 violet	Primary buttons, links, active states, price
<code>--accent</code>	#ff9f1c amber	Product tags and star ratings
<code>--mint</code>	#0f8f86 teal	Supporting highlights and value icons
<code>--rose</code>	#c2255c pink	Wishlist hearts and basket counter
<code>--ink</code> / <code>--ink-soft</code>	#17161c / #56545f	Body text and secondary text
<code>--bg</code> / <code>--surface</code>	#faf8f5 / #ffffff	Page background and cards
<code>--font-display</code>	Fredoka (Google Fonts)	Headings, buttons, prices
<code>--font-body</code>	Plus Jakarta Sans (Google Fonts)	Body copy, forms, tables

The same tokens are redefined once under `[data-theme="dark"]`, which is how the theme switch in the header recolours the entire site without duplicating a single rule. Both themes were checked for contrast and both pass at AA.

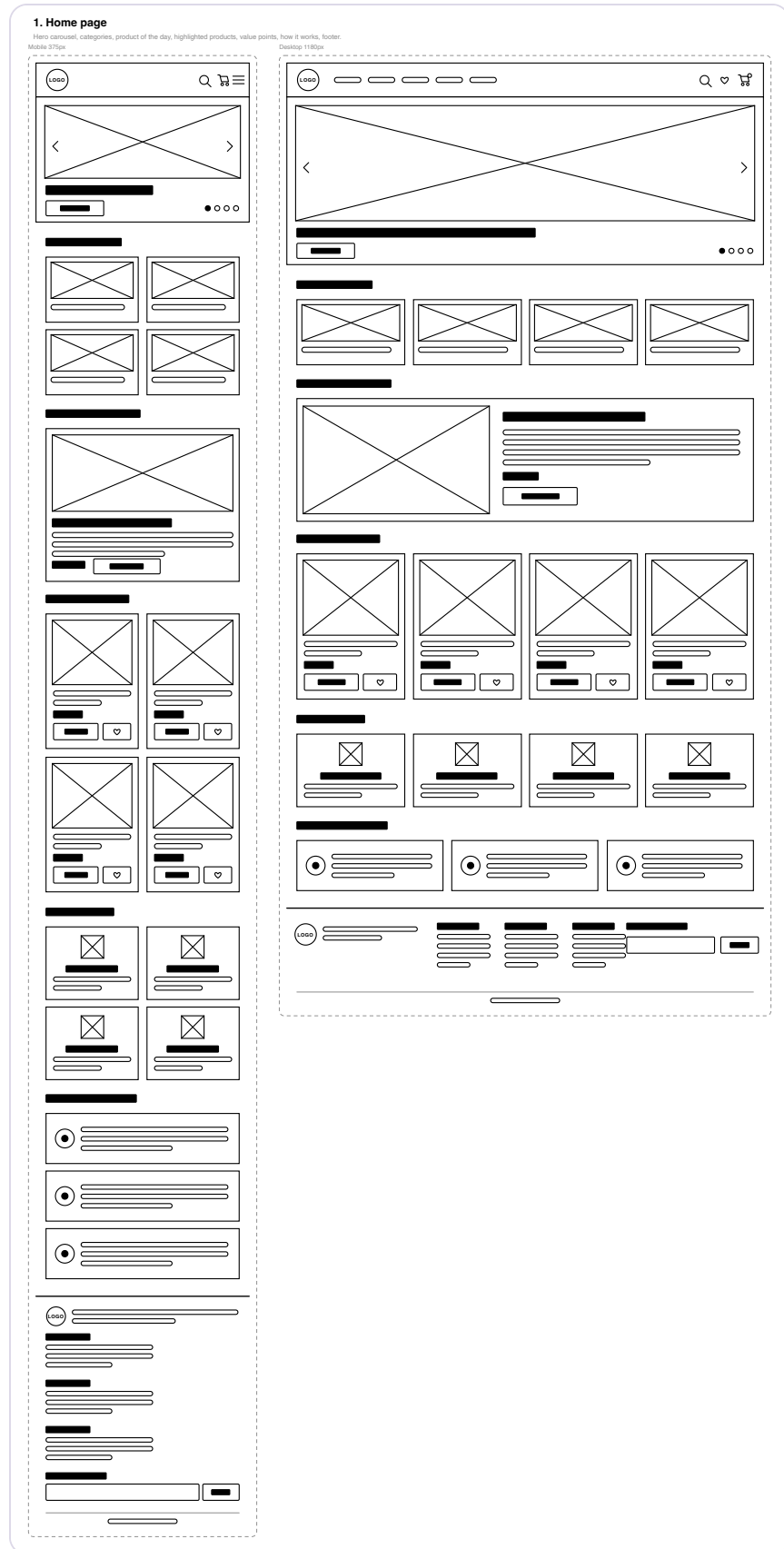
2.3 Wireframes

Each wireframe shows the mobile layout that was designed first next to the desktop layout it grows into. They follow the low-fidelity conventions taught in the module: black and white lines only, no graphics, no dummy text and no colour, so the discussion stays on structure, hierarchy and functionality rather than on visual design.

The notation used throughout is:

- A **rounded outline bar** is one line of text; a few stacked inside a box are a block of copy.
- A **solid black bar** is a heading or an emphasised label such as a price.
- A **box crossed corner to corner** is an image or icon placeholder.
- A **circle marked LOGO** is the brand mark, the only word that appears in the drawings.
- An **empty box** is an input field; a **box with a solid bar inside it** is a button.

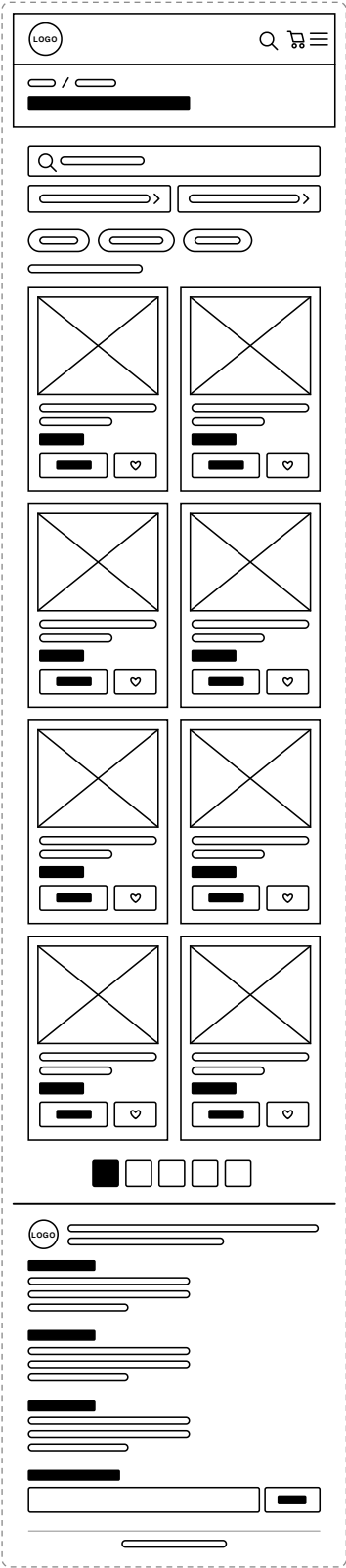
- A **small square** is a checkbox and a **small circle** is a radio button; plus and minus signs are the quantity stepper.



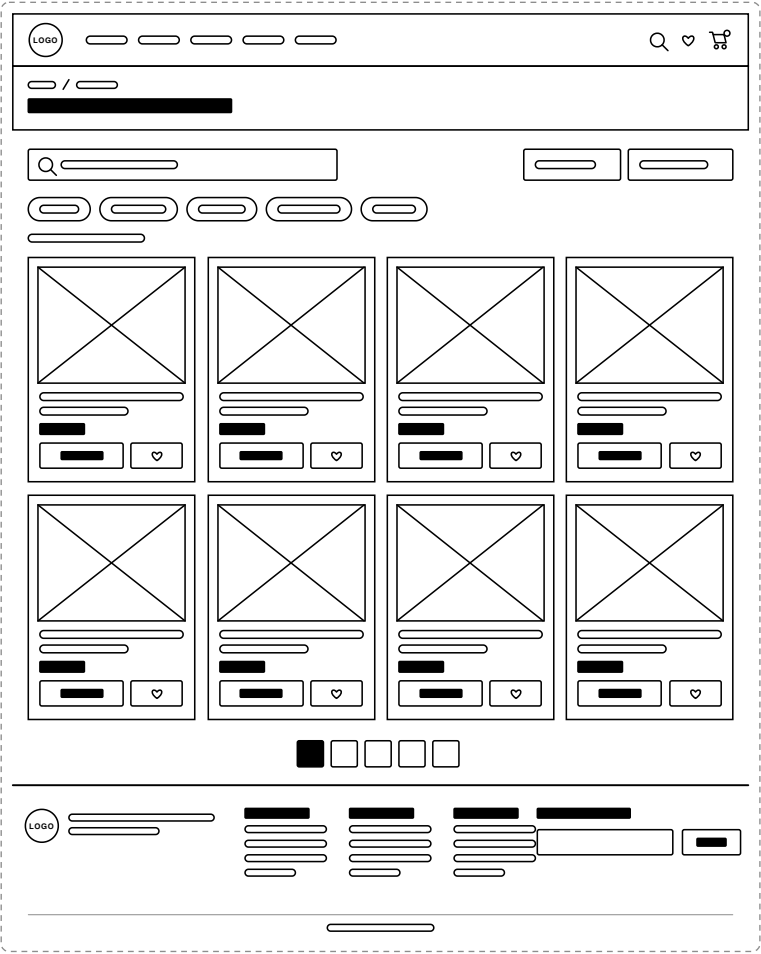
2. Product listing page

Search, sort and category controls, filter chips, product grid and pagination.

Mobile 375px



Desktop 1180px

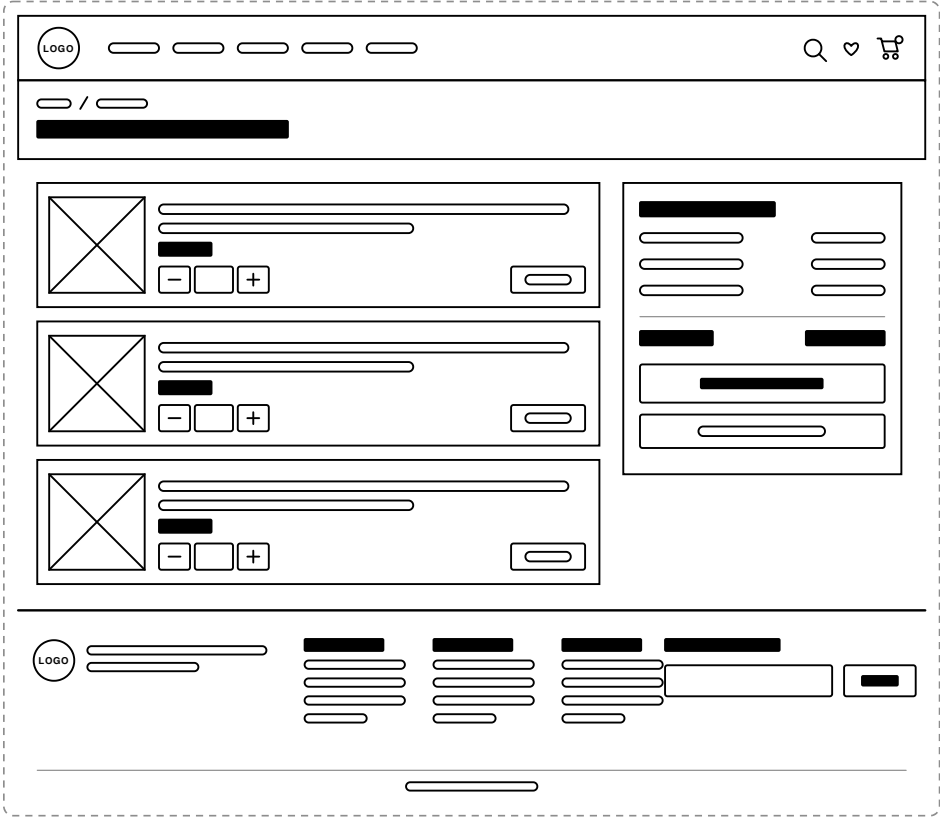
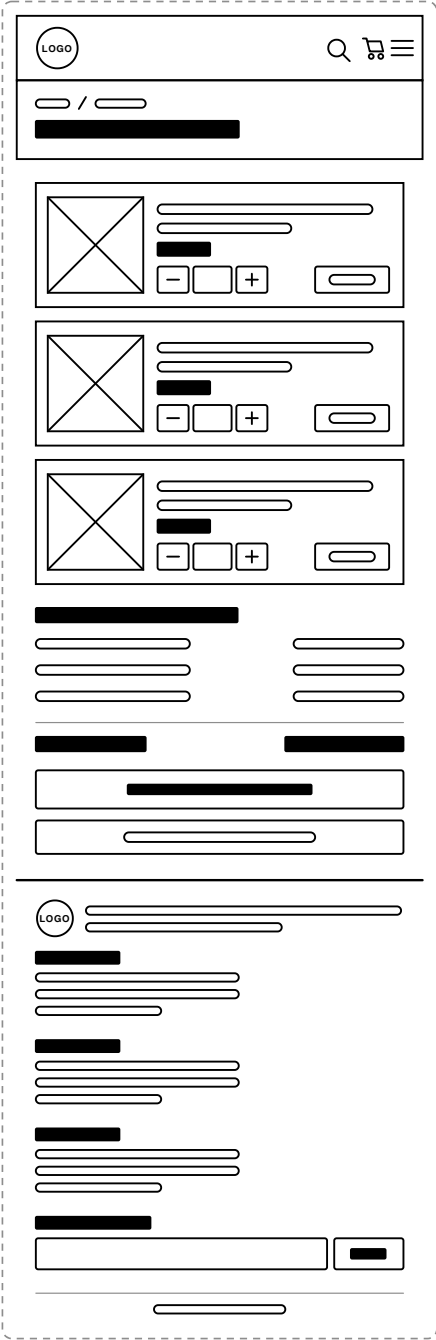


3. Shopping cart page

Cart rows with quantity steppers and remove controls beside the order summary.

Mobile 375px

Desktop 1180px



4. Checkout page

Progress steps, grouped form fields, payment choice and terms beside a persistent order summary.

Mobile 375px

LOGO

Q

Shopping Cart

Address

Progress steps

Form fields

Payment choice

Terms

Order summary

Desktop 1180px

LOGO

Progress steps

Form fields

Payment choice

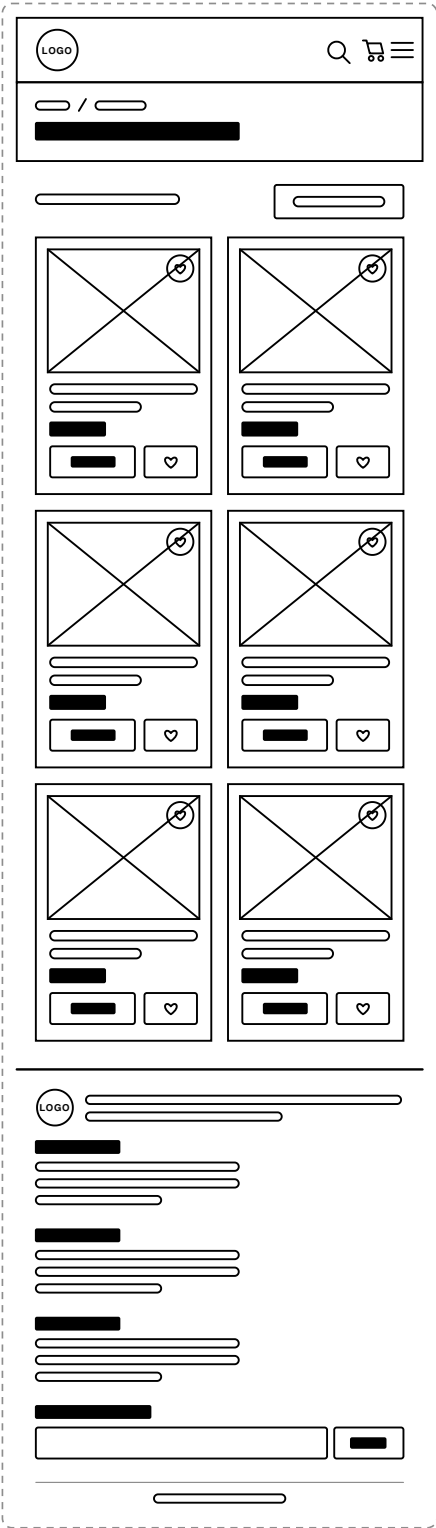
Terms

Order summary

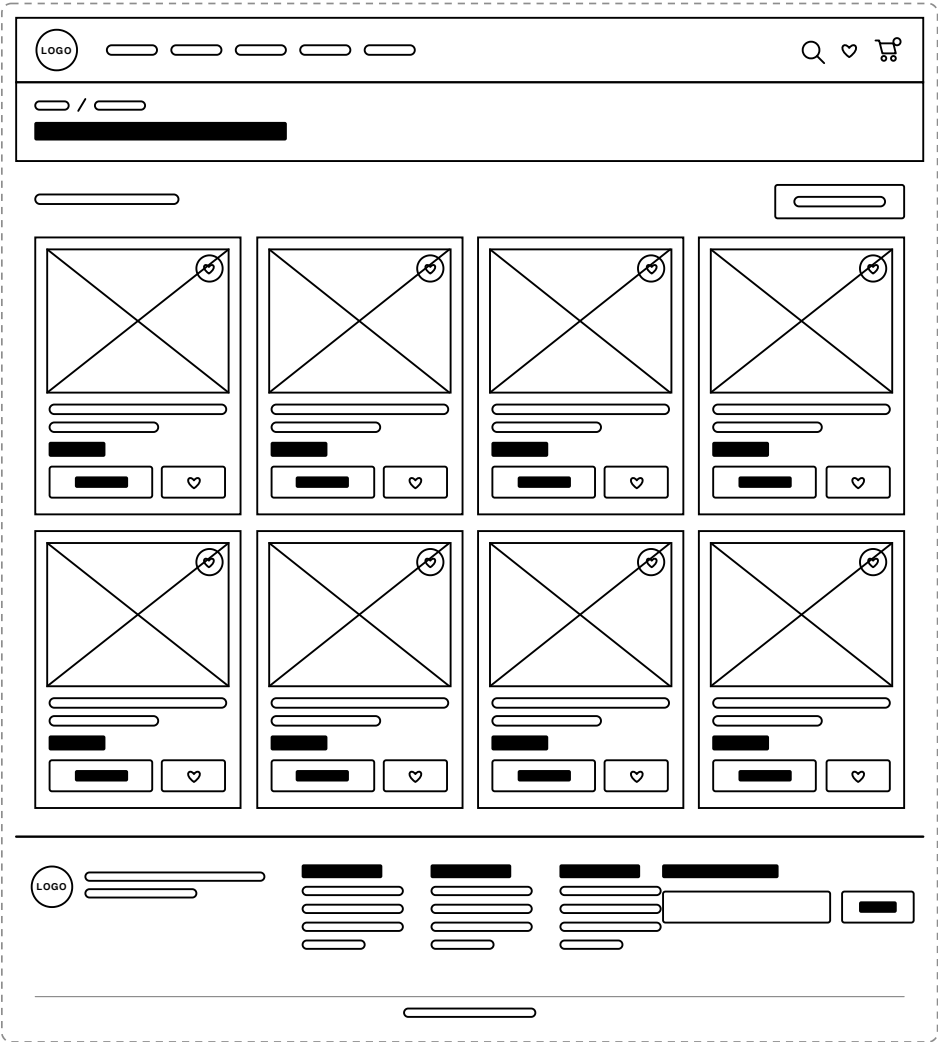
5. Wishlist page

Saved product cards, each with a remove control and an add to cart control.

Mobile 375px



Desktop 1180px



6. Feedback and support page

Contact form and contact details panel above the frequently asked questions accordion.

Mobile 375px

Desktop 1180px

3. Features by page

What each page contains and which requirement it satisfies

Page	Features
Home	Auto-rotating hero carousel (four promotional banners, 6-second interval, pauses on hover, focus and tab change, arrow-key support, dots and prev/next controls); responsive navigation with an animated hamburger; four category cards; Featured Product of the Day chosen from the calendar date; four top-rated product highlights; value grid; an ordered "how it works" list; live statistics calculated from the catalogue; footer newsletter form saved to <code>localStorage</code> .
Products	All 24 products rendered from the JSON catalogue as interactive cards (image, name, category, brand, rating, price); category filter chips with live counts; search by name, brand or category; five sort orders; a shareable query string (<code>?category=toys&q=rocket</code>); a detail modal with a full specification table; Add to Cart and Add to Wishlist on every card; friendly empty state when nothing matches.
Basket	Table of line items with product image, name, unit price, quantity stepper, line subtotal and remove; order summary with subtotal, delivery (free over £60, otherwise £4.95), 5% VAT and total; a note showing how much more is needed for free delivery; clear cart; progress indicator; empty state; the basket persists in <code>localStorage</code> .
Checkout	Validated form (name, email, phone, address, town, postcode); payment method switch between simulated card and cash on delivery, with card fields shown, formatted and validated only when card is selected; confirmation checkbox; live order summary; on success an animated tick, confetti, an order reference, a "what you ordered" table, the basket cleared and the order written to a local order history table.
Wishlist	Everything saved with the heart button, each item markable as Interested , Owned or Not interested ; four live counters; status filter chips; move to basket; clear wishlist; empty states for both "nothing saved" and "nothing in this status".
Support	Fully validated feedback form (name, email, topic, message with a 10–600 character rule); animated confirmation with a reference; a table of messages stored on the device; contact details; an opening-hours table; a six-question FAQ accordion built with JavaScript, one answer open at a time, with arrow-key navigation.
Every page	Skip link, semantic landmarks, sticky header with live basket and wishlist counters, dark/light theme switch, scroll-reveal animations that respect <code>prefers-reduced-motion</code> , toast notifications instead of <code>alert()</code> , favicon set, PWA manifest and service worker.

4. JavaScript architecture and reusable functions

`js/app.js` returns one object, `TH`, that every page uses; `js/components.js` returns `THUI`, which owns the shared markup. Nothing that appears on more than one page is written twice.

Shared functions and where they are reused

Function	Purpose	Used on
<code>TH.formatCurrency()</code>	Formats every price the site shows	All six pages
<code>TH.read()</code> / <code>TH.write()</code>	Safe JSON read and write to <code>localStorage</code> with error handling	All six pages
<code>TH.addToCart()</code> , <code>TH.setQty()</code> , <code>TH.cartTotals()</code>	The single cart API — totals, delivery and VAT are calculated in one place	Home, Products, Basket, Checkout, Wishlist
<code>TH.toggleWishlist()</code> , <code>TH.setWishStatus()</code>	Wishlist storage and status handling	Home, Products, Wishlist
<code>TH.validateField()</code> , <code>TH.validateForm()</code> , <code>TH.wireLiveValidation()</code>	Rule-driven validation that paints inline errors and re-checks on blur	Newsletter (all pages), Checkout, Support
<code>TH.toast()</code>	Custom accessible notifications used instead of <code>alert()</code>	All six pages
<code>TH.getProductOfTheDay()</code>	Picks the daily featured product from the calendar day	Home
<code>THUI.productCard()</code>	Renders one product card — the same markup everywhere a product appears	Home, Products, Wishlist
<code>THUI.openProduct()</code>	Opens the detail modal, traps focus and restores it on close	Home, Products, Wishlist
<code>THUI.emptyState()</code>	Builds the illustrated empty state	Products, Basket, Wishlist

5. Testing

5.1 How the site was tested

- **Functional testing** — the 48 test cases below were executed in Google Chrome 152 on macOS, against the deployed site and a local server.
- **HTML validity** — every page submitted to the W3C Nu Markup Validation Service.
- **CSS validity** — all three stylesheets submitted to the W3C CSS Validation Service (CSS level 3 + SVG profile).
- **Accessibility** — WAVE and axe-core run on every page, in both light and dark themes, including the open modal and the invalid-form state.
- **Lighthouse** — every page audited in both the mobile and desktop presets.
- **Responsiveness** — checked at 320, 375, 414, 768, 1024, 1280 and 1600px.

5.2 Test cases

48 functional test cases — all executed and passed

ID	Area	Test	Steps	Expected result	Status
TC-01	Global	Navigation links	Click each link in the header on every page	The matching page loads and its link is marked as the current page	Pass
TC-02	Global	Animated hamburger menu	Narrow the window below 880px, click the hamburger menu	The bars animate into a cross, the menu slides open, Escape and choosing a link close it again	Pass
TC-03	Global	Skip link	Press Tab on page load, then Enter	The skip link becomes visible and focus jumps to the main content	Pass
TC-04	Global	Theme switch	Click the moon/sun button, reload the page	The site switches theme immediately and the choice is remembered on the next visit	Pass
TC-05	Global	Basket counter	Add any product to the basket	The header counter increases, animates once and matches the basket contents on every page	Pass
TC-06	Global	Wishlist counter	Save a product with the heart button	The wishlist counter in the navigation increases on every page	Pass
TC-07	Global	Footer newsletter — invalid	Enter "not-an-email" and submit	Submission is blocked and "Enter an email address such as you@example.com." appears under the field	Pass

ID	Area	Test	Steps	Expected result	Status
TC-08	Global	Footer newsletter — valid	Enter collector@example.com and submit	A confirmation appears, a toast is shown and the address is stored under toyhaven.newsletter	Pass
TC-09	Global	Footer newsletter — duplicate	Submit the same address twice	The second attempt reports that the address is already subscribed and is not stored twice	Pass
TC-10	Global	Scroll reveal	Scroll down any page	Sections fade and rise into place once, and do not animate when the operating system requests reduced motion	Pass
TC-11	Home	Carousel auto-rotation	Load the home page and wait	The hero changes slide every six seconds and loops back to the first slide	Pass
TC-12	Home	Carousel controls	Click the next, previous and dot controls	The matching slide is shown, the active dot widens and the timer restarts	Pass
TC-13	Home	Carousel pause	Hover over the hero, then move away	Rotation pauses while hovered or focused and resumes afterwards	Pass
TC-14	Home	Carousel keyboard	Focus the hero and press the left and right arrow keys	The previous and next slides are shown	Pass
TC-15	Home	Hidden slides are not focusable	Tab through the hero	Only the links of the visible slide receive focus	Pass
TC-16	Home	Featured Product of the Day	Load the home page	One product is shown with its rating, three specifications, price and buttons; the same product is shown all day	Pass
TC-17	Home	Featured highlights	Load the home page	The four highest-rated products are rendered as cards from the catalogue	Pass
TC-18	Home	Live statistics	Load the home page	Products, categories, average rating and total reviews are calculated from the data, not hard coded	Pass

ID	Area	Test	Steps	Expected result	Status
TC-19	Home	Category cards	Click a category card	The product page opens filtered to that category	Pass
TC-20	Products	Full catalogue renders	Open the product page	24 cards are shown, each with image, name, category, rating, brand, age guidance and price	Pass
TC-21	Products	Category filter	Click "Board Games"	Only the six board games remain and the results line reports "Showing 6 of 24 products in Board Games"	Pass
TC-22	Products	Search	Type "rocket" in the search box	Only Cosmo Launch Rocket remains, updating as you type	Pass
TC-23	Products	Search with no results	Type "zzzz"	The grid empties and an empty state offers to reset the shelf	Pass
TC-24	Products	Sorting	Choose "Price: low to high", then "Customer rating"	Cards reorder correctly for each option	Pass
TC-25	Products	Deep link	Open products.html?category=toys&q=rocket	The page opens already filtered and searched, with the chip and search box pre-set	Pass
TC-26	Products	Reset filters	Apply filters, then click Reset filters	All 24 products return, the search box empties and the query string is cleared	Pass
TC-27	Products	Product modal	Click Details on any card	A modal opens with a large image, description, price and a specification table	Pass
TC-28	Products	Modal close and focus	Press Escape, click the backdrop, click the close button	The modal closes each way and focus returns to the button that opened it	Pass
TC-29	Products	Add to cart from a card	Click Add to Cart	A success toast appears and the header counter increases	Pass
TC-30	Products	Add to wishlist from a card	Click the heart	The heart fills, its label changes to "Remove ..." and the wishlist counter increases	Pass

ID	Area	Test	Steps	Expected result	Status
TC-31	Basket	Line items	Add three different products and open the basket	Each product appears once with image, unit price, quantity and line subtotal	Pass
TC-32	Basket	Increase quantity	Click + on a line	Quantity, line subtotal and all totals increase immediately	Pass
TC-33	Basket	Decrease to zero	Click – until the quantity reaches zero	The line is removed from the basket and the totals update	Pass
TC-34	Basket	Remove	Click Remove on a line	The line disappears, a toast confirms it and the counter updates	Pass
TC-35	Basket	Delivery threshold	Basket under £60, then over £60	Delivery is £4.95 with a note about the shortfall, then Free with a confirmation note	Pass
TC-36	Basket	VAT and total	Compare the summary against a manual calculation	VAT is 5% of the subtotal and the total is subtotal + delivery + VAT	Pass
TC-37	Basket	Persistence	Add products, close the browser, reopen the site	The basket is exactly as it was left	Pass
TC-38	Basket	Clear cart	Click Clear cart	Every line is removed and the empty state with a Shop Now button is shown	Pass
TC-39	Checkout	Empty basket	Open the checkout with an empty basket	The form is replaced by a message explaining there is nothing to check out	Pass
TC-40	Checkout	Required fields	Submit the empty form	Ten inline errors appear, focus moves to the first invalid field and nothing is submitted	Pass
TC-41	Checkout	Format rules	Enter "abc" as the email, "12" as the card number, "99/99" as the expiry	Each field reports its own specific message and blocks submission	Pass
TC-42	Checkout	Cash on delivery	Choose Cash on delivery	Card fields are hidden and are no longer required; the order can be placed without them	Pass

ID	Area	Test	Steps	Expected result	Status
TC-43	Checkout	Order confirmation	Complete the form and submit	An animated tick, confetti, an order reference and a table of what was ordered are shown	Pass
TC-44	Checkout	Basket cleared and history saved	Check the header and the history table after ordering	The basket counter returns to zero and the order appears in the local order history	Pass
TC-45	Wishlist	Status marking	Mark a saved product Interested, then Owned	The chosen button is highlighted, the counters update and the choice survives a reload	Pass
TC-46	Wishlist	Status filters and empty state	Filter by Owned with nothing owned	An empty state explains that nothing is marked owned yet	Pass
TC-47	Support	Feedback validation and storage	Submit an empty form, then a valid one	Errors are shown first; the valid message produces a reference, a confirmation panel and a row in the saved-messages table	Pass
TC-48	Support	FAQ accordion	Click several questions, then use the arrow keys	Only one answer is open at a time, the plus icon rotates and the arrow keys move between questions	Pass

5.3 HTML validation — W3C Nu Markup Validation Service

Every page returns no errors and no warnings

Page	Errors	Warnings	Result
index.html	0	0	Pass
products.html	0	0	Pass
cart.html	0	0	Pass
checkout.html	0	0	Pass
wishlist.html	0	0	Pass
support.html	0	0	Pass
offline.html	0	0	Pass

Two issues were found and fixed during validation: an `<aside>` nested inside a section was changed to a titled `<section>`, and the deprecated space-separated `clip: rect(0 0 0 0)` shorthand in the screen-reader utility was rewritten with commas and paired with `clip-path`.

5.4 CSS validation — W3C CSS Validation Service

CSS level 3 + SVG profile

Stylesheet	Errors	Warnings	Result
css/base.css	0	5 — custom properties cannot be checked statically, plus one vendor prefix	Pass
css/components.css	0	151 — all "CSS variables are not statically checked"	Pass
css/pages.css	0	114 — all "CSS variables are not statically checked"	Pass

The warnings are informational: the validator cannot resolve the value of a CSS custom property at validation time, so it reports one warning for each use. The only other warning is the `-webkit-text-size-adjust` vendor prefix, which is kept deliberately to stop iOS Safari inflating text in landscape.

5.5 Accessibility — WAVE and axe-core

Errors and contrast errors per page, in both themes

Page	Errors	Contrast errors	Result
Home	0	0	Pass
Products (including the open modal)	0	0	Pass
Basket	0	0	Pass
Checkout (including the invalid form state)	0	0	Pass
Wishlist	0	0	Pass
Support (including an open FAQ answer)	0	0	Pass

Accessibility work that produced those results:

- Landmarks on every page: `header`, `nav`, `main`, `footer`, plus a skip link.
- A visible label for every form control, errors announced with `role="alert"` and linked by `aria-invalid`.

- Descriptive alternative text on product images, and empty `alt` on the purely decorative hero backgrounds.
- State exposed to assistive technology: `aria-pressed` on filter, heart and status buttons, `aria-expanded` on the hamburger and FAQ triggers, `aria-current` on the active page and slide.
- The carousel marks hidden slides with `aria-hidden` and removes their links from the tab order.
- The modal sets `aria-modal`, moves focus to its close button and returns focus to the trigger.
- Toasts are announced politely through a live region instead of interrupting with `alert()`.
- All animation is disabled under `prefers-reduced-motion`, and both themes meet AA contrast.

5.6 Lighthouse

Google Lighthouse 12, headless Chrome, default mobile preset and desktop preset

Page	Mobile				Desktop			
	Perf.	Access.	Best pr.	SEO	Perf.	Access.	Best pr.	SEO
Home	99	100	100	100	100	100	100	100
Products	100	100	100	100	91	100	100	100
Basket	100	100	100	100	100	100	100	100
Checkout	100	100	100	100	100	100	100	100
Wishlist	99	100	100	100	100	100	100	100
Support	99	100	100	100	100	100	100	100

All audits were run with Lighthouse 12.8 in headless Chrome. The first audit scored 84 on mobile.

Three changes brought it to 99–100:

1. Google Fonts are now preloaded and applied without blocking the first paint, with a `<noscript>` fallback — this removed 1.75 seconds of render blocking.
2. Every product and banner image ships in two sizes (320px and 640px products, 800px and 1280px banners) served through `srcset` and `sizes`.
3. The first hero image is preloaded and marked `fetchpriority="high"`, which fixed a slow largest-contentful-paint.

The deployed site on GitHub Pages, audited over the public internet

Preset	Performance	Accessibility	Best practices	SEO
Mobile	93	100	100	100

Preset	Performance	Accessibility	Best practices	SEO
Desktop	99	100	100	100

Live performance is a few points lower than the local audit purely because of network latency: Lighthouse measured a 900 ms server response for the first request and GitHub Pages sets a ten-minute cache lifetime, neither of which can be changed from the site's code. Total blocking time is 0 ms and cumulative layout shift is 0.029 on both presets.

5.7 Responsiveness

Layout checked across device widths

Width	Representative device	Behaviour	Result
320px	iPhone SE (1st gen)	Single column, hamburger menu, one card per row, no horizontal scrolling	Pass
375px	iPhone SE / 13 mini	The reference mobile layout; hero copy panel fits without clipping	Pass
414px	iPhone Pro Max	As 375px with more comfortable spacing	Pass
768px	iPad portrait	Two-column cards, footer becomes four columns, modal becomes two columns	Pass
1024px	iPad landscape / small laptop	Full navigation returns, basket and checkout become two columns, four cards per row	Pass
1280–1600px	Desktop	Content is capped at 1180px and centred, so lines stay readable	Pass

5.8 Progressive web app

Installability and offline behaviour

Check	Expected	Result
Manifest	Name, short name, description, scope, start URL, standalone display, theme colours, 192px, 512px and maskable icons, three shortcuts	Pass
Favicons	<code>favicon.ico</code> , 32px PNG and a 180px Apple touch icon on every page	Pass
Service worker	Registers, activates and pre-caches 32 files of the application shell	Pass

Check	Expected	Result
Offline	Visited pages open with the network disabled; an unvisited page falls back to <code>offline.html</code>	Pass
Install	Chrome offers "Install Toy Haven" and the app opens in its own window	Pass

6. Hosting and deployment

1. The project was committed to a dedicated repository named `toy-haven`.
2. GitHub Pages was enabled in *Settings* → *Pages*, deploying from the `main` branch, `/` (root) folder.
3. The site is publicly available at <https://nihmathullahmazhar.github.io/toy-haven/> and was re-tested after deployment.

All paths are relative, so the site works both at the repository sub-path on GitHub Pages and when served from the root of any other server.

7. Known limitations and future work

- There is no back end, so orders, messages and subscriptions stay on the device that created them; clearing browser data clears them.
- Payment is simulated. No card details are transmitted, stored or validated against a payment provider beyond format checks.
- The catalogue is fixed at 24 products in `js/data.js`. It is deliberately shaped like an API response so it could be replaced with a `fetch()` call without touching the rendering code.
- Product photography is replaced by original vector-style illustrations generated for this project, so no third-party images are used.
- Possible next steps: pagination or infinite scroll for a larger catalogue, product reviews, and a background-sync queue so orders placed offline are sent when the connection returns.